Current approaches to modelling the **spread** of **wildland fire**: a **review**

GLW Perry - Progress in Physical Geography, 1998 - journals.sagepub.com

… **simulation** of the spatiotemporal behaviour of **wildland fire** is … risk evaluation and fire **spread simulation** is discussed. Two … of a `new generation' of fire **spread** models; current trends in …

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Machine learning and deep learning for **wildfire spread prediction**: A **review**

HS Andrianarivony, MA Akhloufi - Fire, 2024 - mdpi.com

… highly performant in predicting fire **behavior**. This paper presents a systematic **review** of recent ML and DL techniques developed for **wildfire spread prediction**, detailing the commonly …

☆ Save 99 Cite Cited by 16 Related articles All 4 versions »

Trending and emerging prospects of physics-based and ML-based **wildfire spread** models: A comprehensive **review**

H Singh, LM Ang, T Lewis, D Paudyal, M Acuna… - Journal of Forestry …, 2024 - Springer

… aiding in the development of more accurate and reliable **prediction** tools. Methods for literature **review** This **wildfire spread model review** analysed research published in peer-reviewed …

☆ Save 99 Cite Cited by 25 Related articles All 7 versions

A **review** of genetic algorithm approaches for **wildfire spread prediction** calibration

J Pereira, J Mendes, JSS Júnior, C Viegas, JR Paulo - Mathematics, 2022 - mdpi.com

… **prediction** based on the use of evolutionary algorithms for input parameter set calibration. In … literature **review**, it was observed that the current literature on **wildfire spread prediction** …

☆ Save 99 Cite Cited by 52 Related articles All 8 versions »

A systematic **review** and bibliometric analysis of **wildland fire behavior** modeling

J Silva, J Marques, I Gonçalves, R Brito, S Teixeira… - Fluids, 2022 - mdpi.com

… a mathematical percept to **model** the **spread** of **wildland fire**. Bakhshaii and Johnson [7], in a recent **review**, called these last two models a first-generation type of **wildfire** models. The …

☆ Save 99 Cite Cited by 21 Related articles All 6 versions »

A comparative **review** on **wildfire** simulators

GD Papadopoulos, FN Pavlidou - IEEE systems Journal, 2011 - ieeexplore.ieee.org

… This project promises to develop more reliable **bushfire spread** simulation and animation … In this paper, a **review** on **wildfire simulation** models has been presented. Before that, a brief …

☆ Save 99 Cite Cited by 171 Related articles All 9 versions

A **review** of **wildland fire spread** modelling, 1990-present 3: Mathematical analogues and **simulation** models

AL Sullivan - arXiv preprint arXiv:0706.4130, 2007 - arxiv.org

… of **review** papers endeavours to critically and comprehensively **review** all types of surface fire **spread** … This paper reviews models of a **simulation** or mathematical analogue nature. Most …

☆ Save 99 Cite Cited by 50 Related articles All 6 versions »

A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and Machine Learning Techniques used for the **Prediction** of **Wildfire** in Australia

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… better anticipate and mitigate the impacts of **wildfire spread**. This **review** will discuss the range of empirical and dynamic tools used for **wildfire prediction**, highlighting their advantages, …

☆ Save 99 Cite Cited by 2 Related articles All 4 versions

A **review** of a new generation of **wildfire**–atmosphere modeling

A Bakhshaii, EA Johnson - Canadian Journal of Forest …, 2019 - cdnsciencepub.com

… The earliest physical science models tend to deal with isolated microscale phenomena (eg, **wildfire spread model** or the optical properties of ice crystals) so that their direct application …

☆ Save 99 Cite Cited by 108 Related articles All 12 versions

Facing the **wildfire spread** risk challenge: where are we now and where are we going?

J Sun, W Qi, Y Huang, C Xu, W Yang - Fire, 2023 - mdpi.com

… Consequently, it is imperative to comprehensively **review** and explore further … for **wildfire spread simulation**. In Figure 6, various visual elements of **wildfire spread behavior simulation** …

☆ Save 99 Cite Cited by 23 Related articles All 6 versions »

Uncertainty and risk in **wildland fire** management: a **review**

MP Thompson, DE Calkin - Journal of environmental management, 2011 - Elsevier

… We **review** the state of **wildfire** risk assessment and … Recent advances in **wildfire simulation** and geospatial mapping of … factors contributing to **wildfire** occurrence and **spread**. Managing …

☆ Save 99 Cite Cited by 381 Related articles All 15 versions

A **review** of machine learning applications in **wildfire** science and management

P Jain, SCP Coogan, SG Subramanian… - Environmental …, 2020 - cdnsciencepub.com

… task of modeling **wildland fire spread**, the relevant physical properties include fuel composition, local weather, and topography. The current state-of-the-art method in **wildfire prediction** …

☆ Save 99 Cite Cited by 776 Related articles All 21 versions

A **review** of **wildland fire spread** modelling, 1990-present, 1: Physical and quasi-physical models

AL Sullivan - arXiv preprint arXiv:0706.3074, 2007 - arxiv.org

… Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial scales, IUSTI concentrates on resolving the chemical and conservation equations at a …

☆ Save 99 Cite Cited by 62 Related articles All 4 versions »

Wildland surface fire **spread** modelling, 1990–2007. 3: **Simulation** and mathematical analogue models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… The present series of articles endeavours to comprehensively **survey** and précis all types of surface fire **spread** models developed during the period 1990–2007. The present paper …

☆ Save 99 Cite Cited by 458 Related articles All 5 versions

Research Trends in **Wildland Fire prediction** amidst climate change: A Comprehensive bibliometric analysis

M Bao, J Liu, H Ren, S Liu, C Ren, C Chen, J Liu - Forests, 2024 - mdpi.com

… , we can enhance our understanding of **wildland fire behavior** and improve our ability to … The objective of this study is to conduct a critical thematic **review** of **wildland fire prediction** …

☆ Save 99 Cite Cited by 13 Related articles All 6 versions »

Mathematical models and calculation systems for the study of **wildland fire** behaviour

E Pastor, L Zárate, E Planas, J Arnaldos - Progress in Energy and …, 2003 - Elsevier

This is a **review** of the most important work in **wildland fire** mathematical modelling which … **wildland fire** models to be sorted. Surface fire **spread** models, crown fire initiation and **spread** …

☆ Save 99 Cite Cited by 542 Related articles All 10 versions

Limitations on the accuracy of **model** predictions of **wildland fire** behaviour: a state-of-the-knowledge overview

ME Alexander, MG Cruz - The forestry chronicle, 2013 - pubs.cif-ifc.org

… **model** predictions of **wildland fire** behaviour and includes a short historical overview of **wildland fire** behaviour research as it relates to **model** … Estimating **wildland fire** rate of **spread** in a …

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A **review** of **wildland fire spread** modelling, 1990-present 2: Empirical and quasi-empirical models

AL Sullivan - arXiv preprint arXiv:0706.4128, 2007 - arxiv.org

… to **model** the **spread** and behaviour of wildland fires across the landscape. This series … **review** papers endeavours to critically and comprehensively **review** all types of surface fire **spread** …

☆ Save 99 Cite Cited by 42 Related articles All 4 versions »

Application of deep learning in **forest fire prediction**: a systematic **review**

C Mambile, S Kaijage, J Leo - IEEE Access, 2024 - ieeexplore.ieee.org

… included datasets from British Columbia, Alaska, and Nova Scotia to develop FirePred for a hybrid multitemporal convolutional neural network **model** for **wildfire spread prediction**. The …

☆ Save 99 Cite Cited by 13 Related articles All 3 versions

Evacuation **simulation** under threat of **wildfire**—an overview of research, development, and knowledge gaps

SM Beyki, A Santiago, L Laím, HD Craveiro - Applied Sciences, 2023 - mdpi.com

… **review** paper on **wildfire** evacuation modeling involved a systematic process of data collection and literature **review**… integration of **wildfire behavior** into the evacuation **simulation** process…

☆ Save 99 Cite Cited by 15 Related articles All 7 versions »

Simulation of **wildfire** evacuation with dynamic factors and **model** composition

A Beloglazov, M Almashor, E Abebe, J Richter… - **Simulation** Modelling …, 2016 - Elsevier

… **simulation** systems are essential tools for such planning and decision making. During a **wildfire** … Our approach composes several **simulation** and modelling systems, including a **wildfire** …

☆ Save 99 Cite Cited by 113 Related articles All 4 versions

Operational **wildfire** suppression modelling: a **review** evaluating development, state of the art and future directions

TJ Duff, KG Tolhurst - International Journal of **Wildland Fire**, 2015 - CSIRO Publishing

… **review**, we focus on the development and use of models that emulate fire suppression for decision support. We **review** … There are two main approaches to dynamic fire **spread simulation**…

☆ Save 99 Cite Cited by 91 Related articles All 7 versions

Evaluating a **model** for predicting active crown fire rate of **spread** using **wildfire** observations

ME Alexander, MG Cruz - Canadian Journal of Forest Research, 2006 - cdnsciencepub.com

… With 57 quantitative **wildfire** observations, no study to date … to evaluate a predictive fire **behavior model**. Thus, contrary to … — A **review** of fire weather, fuels, and fire **behavior**. Can. For. …

☆ Save 99 Cite Cited by 146 Related articles All 14 versions

DEVS-FIRE: Towards an integrated **simulation** environment for surface **wildfire spread** and containment

L Ntaimo, X Hu, Y Sun - **Simulation**, 2008 - journals.sagepub.com

… applied to an example of fire **spread simulation** [44]. A recent … the **wildfire spread** and containment **simulation model** presented in … We **review** the basics of **wildfire behavior** [56] to set …

☆ Save 99 Cite Cited by 122 Related articles All 9 versions

Wildland surface fire **spread** modelling, 1990–2007. 1: Physical and quasi-physical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… readers, and this is the main reason for the inclusion of his work in the current **review**. … Whereas FIRETEC was intended to be used to **model wildland fire spread** across large spatial …

☆ Save 99 Cite Cited by 625 Related articles All 5 versions

Wildland fire spread modelling using cellular automata: evolution in large-scale spatially heterogeneous environments under fire suppression tactics

A Alexandridis, L Russo, D Vakalis… - … of **Wildland Fire**, 2011 - CSIRO Publishing

… We also use the proposed **model** to simulate a much bigger **wildfire** where both evolutionary data for the fire front and the fire-suppression tactics were available. The study concerns the …

☆ Save 99 Cite Cited by 100 Related articles All 8 versions

Wildland fire spread modeling using convolutional neural networks

JL Hodges, BY Lattimer - Fire technology, 2019 - Springer

… **spread** requires multiple simulations that may limit the number of scenarios which can be analyzed during an emergency. A new **wildland fire spread model** … **wildland fire spread model** …

☆ Save 99 Cite Cited by 143 Related articles All 5 versions

A data-driven **model** for large **wildfire** behaviour **prediction** in Europe

D Rodriguez-Aseretto, D de Rigo, M Di Leo… - Procedia Computer …, 2013 - Elsevier

… as a required input for a realistic **prediction** of the **wildfire dynamics**. In this case, the fire controls are decided elsewhere and notified to the predictive **model**. This may be exploited by …

☆ Save 99 Cite Cited by 48 Related articles All 12 versions

A **Survey** of **Wildfire Spread Prediction** and Risk Estimation Methodswith Machine Learning Techniques

S Dascalu, R Wu, A Munoz, S Guan - 2022 - scholarsarchive.byu.edu

… we conducted a **survey** on papers published recently focused on **wildfire spread prediction** and … In the rest of the paper, Section 2 illustrates how **wildfire spread** behaviour is predicted in …

☆ Save 99 Cite Related articles »

Modelling smoke transport from wildland fires: a **review**

SL Goodrick, GL Achtemeier, NK Larkin… - … of **Wildland Fire**, 2012 - CSIRO Publishing

… Another grid **model** receiving attention for **wildland fire** related air quality issues is the … of the modelling studies cited in this **review** involves the **dynamics** of the buoyant phase of the …

☆ Save 99 Cite Cited by 147 Related articles All 8 versions

Applications of **simulation**-based burn probability modelling: a **review**

MA Parisien, DA Dawe, C Miller… - … of **wildland fire**, 2019 - CSIRO Publishing

… takes advantage of fire **spread** algorithms created for operational uses and the proliferation of available data representing **wildfire** patterns, fuels and weather. This **review** describes this …

☆ Save 99 Cite Cited by 113 Related articles All 13 versions

Advances in **Wildfire Spread** Detection and **Prediction**: Techniques, Challenges, and Applications

RY Zakari, K Kalinaki, ZK Lawal… - Artificial Intelligence and … - taylorfrancis.com

… DL algorithms in revolutionizing **wildfire** detection and **prediction**, enabling … to **forecast wildfire behavior** and **spread** patterns. … Spatial **prediction** of **wildfire** susceptibility using field **survey** …

☆ Save 99 Cite Related articles »

Wildfire prediction to inform fire management: statistical science challenges

SW Taylor, DG Woolford, CB Dean, DL Martell - 2013 - projecteuclid.org

… In this paper we **review** some of the models that have been developed to … **prediction** error into these sources when evaluating the accuracy of a particular **spread model** against **wildfire** …

☆ Save 99 Cite Cited by 193 Related articles All 13 versions

Wildland surface fire **spread** modelling, 1990–2007. 2: Empirical and quasi-empirical models

AL Sullivan - International Journal of **Wildland Fire**, 2009 - CSIRO Publishing

… **spread** for the prevailing conditions (the **prediction** of which is the primary aim of all fire **spread prediction** … covered in the present **review** as it is outside the scope of surface fire **spread**). …

☆ Save 99 Cite Cited by 579 Related articles All 4 versions

WRF-Fire: coupled weather–**wildland fire** modeling with the weather research and forecasting **model**

JL Coen, M Cameron, J Michalakes… - Journal of Applied …, 2013 - journals.ametsoc.org

… numerical weather **prediction model**. The fire module is a surface fire-**behavior model** that is two-way coupled with the atmospheric **model**. Near-surface winds from the atmospheric …

☆ Save 99 Cite Cited by 337 Related articles All 15 versions

A method for ensemble **wildland fire simulation**

MA Finney, IC Grenfell, CW McHugh, RC Seli… - … Modeling & Assessment, 2011 - Springer

… An ensemble **simulation** system that accounts for uncertainty in long-range weather conditions and two-dimensional **wildland fire spread** is described. Fuel moisture is expressed based …

☆ Save 99 Cite Cited by 266 Related articles All 11 versions

Overview of **bushfire spread simulation** systems

P Johnston, G Milne, D Klemitz - **BUSHFIRE** CRC Project B, 2005 - naturalhazards.com.au

… a **review** of the state of the art of **bushfire simulation** systems. We discuss the input data for a **bushfire** behaviour **model** and … in various **bushfire simulation** systems. We concentrate on six …

☆ Save 99 Cite Cited by 22 Related articles All 3 versions »

Modeling **wildfire spread** in wildland-industrial interfaces using dynamic Bayesian network

N Khakzad - Reliability Engineering & System Safety, 2019 - Elsevier

… **wildfire spread** in WIIs by combining dynamic Bayesian network and **wildfire behavior prediction** … The developed methodology models the spatial and temporal **spread** of fire, based on …

☆ Save 99 Cite Cited by 84 Related articles All 10 versions

Forest fire spread simulating **model** using cellular automaton with extreme learning machine

Z Zheng, W Huang, S Li, Y Zeng - Ecological Modelling, 2017 - Elsevier

… The quantitative **simulation** of **forest fire** spreading plays an … By integrating the ELM with the traditional **forest fire** CA framework… actual fire **behavior** observations shows that its **simulation** …

☆ Save 99 Cite Cited by 130 Related articles All 8 versions

Robust **Prediction** of **Wildfire Spread** in Australia

M Palk, K Knappmann, S Voss, R Jovanovic - IEEE Access, 2025 - ieeexplore.ieee.org

… , a recent **survey** can be found about the application … **wildfire prediction** by conducting a computational study with predictive models to robustly estimate daily and weekly **wildfire spread** …

☆ Save 99 Cite Related articles All 2 versions

Fast-moving dire **wildfire** evacuation **simulation**

S Grajdura, S Borjigin, D Niemeier - Transportation research part D …, 2022 - Elsevier

… There is a gap in our knowledge of human evacuation **behavior** under these conditions. In … **wildfire** evacuation. We use a post-disaster **survey** and decision tree methods to **model** agent …

☆ Save 99 Cite Cited by 45 Related articles All 5 versions

PR0PAGAT0R: An Operational Cellular-Automata Based **Wildfire** Simulator

A Trucchia, M D'Andrea, F Baghino, P Fiorucci… - Fire, 2020 - mdpi.com

… **model** for **forest fire spread simulation**, conceived as a rapid method for fire risk assessment. The **model** … The fire **spread** probability depends on vegetation type, slope, wind direction …

☆ Save 99 Cite Cited by 70 Related articles All 10 versions »

A **forest fire prediction model** based on cellular automata and machine learning

X Sun, N Li, D Chen, G Chen, C Sun, M Shi… - IEEe …, 2024 - ieeexplore.ieee.org

… a **Forest Fire Spread Behavior Prediction** (FFSBP) **model**, encompassing two integral components: the **Forest Fire Spread** Process Prediction (FFSPP) **model** … literature **survey** is that CA …

☆ Save 99 Cite Cited by 20 Related articles All 2 versions

Modeling **wildland fire propagation** with level set methods

V Mallet, DE Keyes, FE Fendell - Computers & Mathematics with …, 2009 - Elsevier

… fire **spread**, one seeks a computationally efficient **model**, … burning, and post-crisis **review**, eg, personnel training, litigation)… treating the **model** formulated to simulate **wildland fire** evolution…

☆ Save 99 Cite Cited by 140 Related articles All 17 versions

Driving **wildfire spread prediction** by multi-source real-time observations

C Lin, Y Shi, Z Wang, M Zha, X Li, J Ji - Ecological Modelling, 2025 - Elsevier

… **Wildfire spread prediction** based on data assimilation (DA) enhances **forecast** accuracy … watchtower monitoring) and **simulation** studies. Results demonstrate significant improved …

☆ Save 99 Cite Related articles

Modeling post-**wildfire** hydrologic response: **Review** and future directions for applications of physically based distributed **simulation**

BA Ebel, ZM Shephard, MA Walvoord… - Earth's …, 2023 - Wiley Online Library

… Building a community **wildfire model** that simulates the links between **wildfire spread**, **wildfire** effects on vegetation and soils, hydrologic and erosion processes, biogeochemical cycling, …

☆ Save 99 Cite Cited by 47 Related articles All 8 versions

An evaluation of NDFD weather forecasts for **wildland fire behavior prediction**

WG Page, NS Wagenbrenner, BW Butler… - Weather and …, 2018 - journals.ametsoc.org

… , 2015: Wildland firefighter safety and fire **behavior prediction** on the fireline. Proc. 13th Int… of Federal **Wildland Fire** Management Policy. National **Wildfire** Coordinating Group Memo. …

☆ Save 99 Cite Cited by 27 Related articles All 7 versions

A cellular automaton **model** of **wildfire propagation** and extinction

KC Clarke, JA Brass, PJ Riggan - … Engineering and Remote …, 1994 - research.fs.usda.gov

… We propose a new **model** to predict the spatial and temporal **behavior** of wildfires. Fire **spread** and intensity were simulated using a cellular automaton **model**. Monte Carlo techniques …

☆ Save 99 Cite Cited by 299 Related articles All 11 versions »

Emulation of **wildland fire spread simulation** using deep learning

F Allaire, V Mallet, JB Filippi - Neural networks, 2021 - Elsevier

… Numerical **simulation** of **wildland fire spread** is useful to … of the burned surface returned after **simulation** of a fire igniting … After training on a large **simulation** dataset, the network shows …

☆ Save 99 Cite Cited by 62 Related articles All 11 versions

Mathematical modeling of **forest fire**-comprehensive **review**

KRA Pillai, YED Sultan - Indian J. Environ. Prot., 2024 - researchgate.net

… 1CFME data is very useful to verify mathematical **model** … CO64), This **review** article aims to present **forest fire** models briefly. … **wild land fire propagation** using semi-physical network **model**…

☆ Save 99 Cite Cited by 1 Related articles

Spatial **prediction** of **wildfire** susceptibility using field **survey** GPS data and machine learning approaches

O Ghorbanzadeh, K Valizadeh Kamran, T Blaschke… - Fire, 2019 - mdpi.com

… A **wildfire** data inventory was created by integrating global …) approaches for the spatial **prediction** of **wildfire** susceptibility. The … All ML approaches were trained using 75% of the **wildfire** …

☆ Save 99 Cite Cited by 173 Related articles All 13 versions »

Physical **model** of **wildland fire spread**: Parametric uncertainty analysis

X Yuan, N Liu, X Xie, DX Viegas - Combustion and Flame, 2020 - Elsevier

… the **prediction** of **spread** rate may arise from uncertain, imprecise or improper determinations of the **model** … is made on a typical **model** of upslope fire **spread** that is consistent with many …

☆ Save 99 Cite Cited by 33 Related articles All 2 versions

Progress and prospects for predicting **wildfire spread** through the wildland-urban interface

O Price, S Ondei, DMJS Bowman - International Journal of Disaster Risk …, 2025 - Elsevier

… Scientific understanding of **wildfire spread** into urban areas is … **review** the state of knowledge around modelling fire **spread** in … We suggest that our proposed **model** is feasible given …

☆ Save 99 Cite Related articles All 2 versions

Prediction of **wildfire** rate of **spread** in grasslands using machine learning methods

S Khanmohammadi, M Arashpour… - … Modelling & Software, 2022 - Elsevier

… **Prediction** of **wildfire propagation** plays a crucial role in reducing … models to predict the rate of **spread** of grassfires. A dataset from … These models produced excellent fits to the **model** …

☆ Save 99 Cite Cited by 44 Related articles All 6 versions

Statistical models of key components of **wildfire** risk

DDZ Xi, SW Taylor, DG Woolford… - … **Review** of Statistics and …, 2019 - annualreviews.org

… It is important to note that not all **wildland fire** ignitions may … fire **spread** may be detected by **wildland fire** management … Improving **prediction** of daily **wildfire dynamics** is critical for …

☆ Save 99 Cite Cited by 72 Related articles All 6 versions

On the need for a theory of **wildland fire spread**

MA Finney, JD Cohen, SS McAllister… - … journal of **wildland fire**, 2012 - CSIRO Publishing

… In summary, our **review** of existing research indicates an insufficient experimental basis for … an insufficient basis for related fire **spread model** assumptions. Independent experimental …

☆ Save 99 Cite Cited by 310 Related articles All 12 versions

Representation and evaluation of **wildfire propagation** simulations

JB Filippi, V Mallet, B Nader - … Journal of **Wildland Fire**, 2013 - CSIRO Publishing

… a **wildfire simulation**, reviews the most common scoring methods using this formalism, and proposes new methods that are explicitly designed to evaluate a **forest fire simulation** … **review** …

☆ Save 99 Cite Cited by 72 Related articles All 11 versions

Wildfire risk modeling

S Oliveira, J Rocha, A Sá - Current Opinion in Environmental Science & …, 2021 - Elsevier

… **review** of the methods and approaches applied to **model wildfire** risk and its key components, in particular since 2018, and how they can contribute to **wildfire** management. …

☆ Save 99 Cite Cited by 76 Related articles All 5 versions

Predicting **wildfire spread** and behaviour in Mediterranean landscapes

M Salis, B Arca, F Alcasena… - … of **Wildland Fire**, 2016 - CSIRO Publishing

… Nevertheless, to incorporate fire **spread simulation** in operational workflows, it is crucial to increase the **model** accuracy by suitable validation. The validation of modelled fire behaviour …

☆ Save 99 Cite Cited by 77 Related articles All 8 versions

Forest fire spread simulation and fire extinguishing visualization research

Q Meng, H Lu, Y Huai, H Xu, S Yang - Forests, 2023 - mdpi.com

… **forest fire spread** quickly. Therefore, this paper proposes a lightweight **forest fire spread model** … ensuring the realism of **forest fire behavior** and the immersion of the virtual environment. …

☆ Save 99 Cite Cited by 17 Related articles All 5 versions »

A **review** of recent advances in risk analysis for **wildfire** management

C Miller, AA Ager - International journal of **wildland fire**, 2012 - CSIRO Publishing

… We also **review** recent advances in addressing temporal **dynamics** of … for future research and development in **wildfire** risk analysis. … **Wildfire simulation** outputs were used to estimate burn …

☆ Save 99 Cite Cited by 344 Related articles All 11 versions

Towards a dynamic data driven **wildfire behavior prediction** system at european level

T Artés, A Cencerrado, A Cortés, T Margalef… - Procedia Computer …, 2014 - Elsevier

… (European **Forest Fire** … the **prediction** of **forest fire spread** is still at an initial stage. This work is a prove of concept to update the system to support near-real time **forest fire spread forecast** …

☆ Save 99 Cite Cited by 17 Related articles All 12 versions

Generative AI as a Pillar for Predicting 2D and 3D **Wildfire Spread**: Beyond Physics-Based Models and Traditional Deep Learning

H Xu, S Zlatanova, R Liang, I Canbulat - Fire, 2025 - mdpi.com

… specific type of generative AI **model** for **wildfire prediction**. In contrast, … AI models to **wildfire**-**spread prediction**, discussing the … **review** of studies applying generative AI to **wildfire** …

☆ Save 99 Cite Related articles All 2 versions »

Wildfire and weather radar: A **review**

N McCarthy, A Guyot, A Dowdy… - Journal of Geophysical …, 2019 - Wiley Online Library

… is determined by interacting processes of **wildfire behavior** and atmospheric convection. The … Advances could be yielded from a generalized microphysics **model** of **wildfire** plumes, as …

☆ Save 99 Cite Cited by 58 Related articles All 7 versions

Bushfire rate-of-**spread** forecasting: Deterministic and statistical approaches to fire modelling

T Beer - Journal of Forecasting, 1991 - Wiley Online Library

… These are the Rothermel **model** and the McArthur meters, … **model** overpredicted **spread** rates while the McArthur **model** underpredicted. To improve the **forecast** of **bushfire** rate of **spread** …

☆ Save 99 Cite Cited by 41 Related articles All 4 versions

Super real-time **forecast** of **wildland fire spread** by a dual-**model** deep learning method

YZ Li, ZL Wang, XY Huang - Journal of environmental …, 2024 - ira.lib.polyu.edu.hk

… : • A **review** of the development of **wildfire** modelling was con… The target of our study is to **forecast** any **wildfire spread** … the prescribed burning or **forecast wildfire** burnt areas more …

☆ Save 99 Cite Cited by 7 Related articles All 5 versions »

SPARK–A **bushfire spread prediction** tool

C Miller, J Hilton, A Sullivan, M Prakash - International Symposium on …, 2015 - Springer

… system for bushfire **spread prediction**, SPARK, which was developed with the functionality to **model** some of these … Submitted to Environmental Modelling and Software (under **review**) …

☆ Save 99 Cite Cited by 127 Related articles All 7 versions

An empirical-based **model** for predicting the forward **spread** rate of wildfires in eucalypt forests

MG Cruz, NP Cheney, JS Gould… - … of **Wildland Fire**, 2021 - CSIRO Publishing

… recent **wildfire** events. We aimed to evaluate the **model** against independent experimental fire and **wildfire** … of **Wildland Fire**, but were blinded from the peer-**review** process for this paper. …

☆ Save 99 Cite Cited by 47 Related articles All 10 versions

A **review** of **wildfire** occurrence research

MP Plucinski - **Bushfire** Cooperative Research Centre …, 2012 - naturalhazards.com.au

Wildfire occurrence statistics describe the presence and quantity of ignitions across spatial and temporal scales. Fire occurrence research is based on regional fire incidence data …

☆ Save 99 Cite Cited by 32 Related articles All 2 versions

Towards predictive data-driven simulations of **wildfire spread**–Part II: Ensemble Kalman Filter for the state estimation of a front-tracking simulator of **wildfire spread**

MC Rochoux, C Emery, S Ricci… - Natural Hazards and …, 2015 - nhess.copernicus.org

… data-driven simulations are capable of correcting inaccurate predictions of the fire front location and of providing an optimized **forecast** of the **wildfire behavior**; the quality of the **forecast** …

☆ Save 99 Cite Cited by 70 Related articles All 14 versions »

Software-based simulations of **wildfire spread** and wind-fire interaction

M Ghodrat, F Shakeriaski, SA Fanaee, A Simeoni - Fire, 2022 - mdpi.com

… this **review** will be on providing an overview of available **wildfire** … The fire **behavior** models including available **simulation** … **review** of the available software used to **model** fire **spread**. The …

☆ Save 99 Cite Cited by 12 Related articles All 5 versions »

Assessing improvements in models used to operationally predict **wildland fire** rate of **spread**

MG Cruz, ME Alexander, AL Sullivan, JS Gould… - … Modelling & Software, 2018 - Elsevier

… The **prediction** of fire **propagation** across landscapes is … in Australia for predicting fire **spread** rates in five different fuel … contrasted **model** predictions against observed **spread** rates of …

☆ Save 99 Cite Cited by 80 Related articles All 5 versions

A **Survey** on Anomaly detection in Evolving Data: [with Application to **Forest Fire** Risk **Prediction**]

M Salehi, L Rashidi - ACM SIGKDD Explorations Newsletter, 2018 - dl.acm.org

… **model** hence the spatiotemporal customisation of the **model** cannot be assessed. In [31], it is claimed that **forest fire** risk needs to consider both **forest fire** … controlling **forest fire spread**, …

☆ Save 99 Cite Cited by 130 Related articles All 7 versions

Predicting fire **propagation** across heterogeneous landscapes using WyoFire: A Monte Carlo-driven **wildfire model**

CW Ott, B Adhikari, SP Alexander, P Hodza, C Xu… - Fire, 2020 - mdpi.com

… of a **wildfire simulation** platform that implements a Monte Carlo-based **wildfire model** called … datasets were acquired from the United States Geological **Survey** (USGS) LANDFIRE …

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An historical **review** of the simplified physical fire **spread model** PhyFire: **Model** and numerical methods

MI Asensio, JM Cascón, D Prieto-Herráez, L Ferragut - Applied Sciences, 2023 - mdpi.com

… A historical **review** is conducted of PhyFire, a simplified physical **forest fire spread model** developed by the research group on Numerical **Simulation** and Scientific Computation (…

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A real-time computer application for the **prediction** of fire **spread** across the Australian landscape

JR Coleman, AL Sullivan - **Simulation**, 1996 - journals.sagepub.com

… Models for the **prediction** of **wildfire spread** and **wildfire** behaviour across landscapes have … we used the empirically derived McArthur **forest fire spread** algorithm, a prototype version of …

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Uncertainty in **model** predictions of **wildland fire** rate of **spread**

MG Cruz, ME Alexander - 2014 - Imprensa da Universidade de …

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A **review** of operations research methods applicable to **wildfire** management

JP Minas, JW Hearne… - … Journal of **Wildland Fire**, 2012 - CSIRO Publishing

… **Simulation** and SD methods can be used to **model** the **dynamics** of complex systems to gain insights into the problem structure and possible management prescriptions through the use …

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Wildland fire behaviour: **dynamics**, principles and processes

MA Finney, SS McAllister, JM Forthofer, TP Grumstrup - 2021 - books.google.com

… **spread model** developed by Richard (Dick) Rothermel and published in 1972. This **model** became the basis for **wildland fire** … In Chapter 9 we **review** methodologies and challenges for …

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Forest fire propagation prediction based on overlapping DDDAS forecasts

T Artes, A Cardil, A Cortés, T Margalef, D Molina… - Procedia Computer …, 2015 - Elsevier

… In this paper, we propose to a Dynamic Data Driven Application System for **forest fire spread prediction** based on overlapping the results of two DDDAS schemes. In the next section, the …

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Predicting **wildfire** induced changes to runoff: A **review** and synthesis of modeling approaches

D Partington, M Thyer, M Shanafield… - Wiley …, 2022 - Wiley Online Library

… of different **model** types, improvements in predicting **wildfire** induced hydrological impacts
… The ET **dynamics** post-**wildfire** is also dependent on the regeneration strategies of different …

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A systematic **review** of applications of machine learning techniques for **wildfire** management decision support

K Bot, JG Borges - Inventions, 2022 - mdpi.com

… The 21 papers addressing this **wildfire** cycle management … : **wildfire** detection, **wildfire spread prediction**, and **wildfire** … to **wildfire** detection (A), 23.8% to **wildfire spread prediction** (B)…

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Wildfire spreading **prediction** using multimodal data and deep neural network approach

D Shadrin, S Illarionova, F Gubanov, K Evteeva… - Scientific reports, 2024 - nature.com

… critical issue in fire **behavior prediction**, so we analyze their contribution to the **model**’s results.
… In this study, our focus is on predicting **wildfire spread** using multimodal data to assist in …

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A general **model** for **wildfire propagation** with wind and slope

M Ángel Javaloyes, E Pendás-Recondo… - SIAM Journal on Applied …, 2023 - SIAM

… In the **model** we will mainly work in these coordinates, passing all the information from the
… the **wildfire propagation** more easily, and then back to \(\hat N\) to see the actual **spread**. …

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A GIS-based fire **spread** simulator integrating a simplified physical **wildland fire model** and a wind field **model**

D Prieto Herráez, MI Asensio Sevilla… - International Journal …, 2017 - Taylor & Francis

… The **wildland fire model** we propose is called Physical **Forest Fire Spread** (PhFFS) (Asensio and … It is a simplified 2D one-phase semi-physical **model** for surface fires based on the …

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Simulating **forest fire spread** and fire-fighting using cellular automata

K Mutthulakshmi, MRE Wee, YCK Wong, JW Lai… - Chinese Journal of …, 2020 - Elsevier

… of wave **propagation** allow an understanding of the natural **spread** of fire, our **model** applies
… strategies, with the spatial and **propagation dynamics** of fire considered. Cellular automata is …

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Analyzing **wildfire** evacuation **dynamics** with agent-based modeling in damaged road networks

F Ma, JY Lee - Safety Science, 2025 - Elsevier

… In this context, **wildfire** evacuation **simulation** can serve as an … **wildfire** evacuations in damaged transportation settings. The proposed framework uniquely integrates **wildfire simulation** …

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Simulating **forest fire spread** with cellular automation driven by a LSTM based speed **model**

X Li, M Zhang, S Zhang, J Liu, S Sun, T Hu, L Sun - Fire, 2022 - mdpi.com

… **forest fire spread** are studied with the accurate **prediction** of ROS and excellent performance of the **propagation** … , and CA is used as the fire **spread model**. The previous wind speed and …

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Systematic **review** and research gaps on **wildfire** evacuations: infrastructure, transportation modes, networks, and planning

SN Zehra, SD Wong - Transportation Planning and Technology, 2024 - Taylor & Francis

… We chose against searching for and including papers on simulations and **behavior** in **wildfire**

… For instance, a 2009 Australian **bushfire** case study applied a rerouting problem **model** to …

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Deciphering the impact of uncertainty on the accuracy of large **wildfire spread** simulations

A Benali, AR Ervilha, ACL Sá, PM Fernandes… - Science of the Total …, 2016 - Elsevier

… assignment and typology, location and timing of ignitions, had a major impact on **prediction**

… fire **spread simulation** approaches, and provide the necessary data for any fire **model** user …

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Emergency logistics for **wildfire** suppression based on forecasted disaster evolution

Z Yang, L Guo, Z Yang - Annals of Operations Research, 2019 - Springer

… fire **propagation model** is first built using both the flame-igniting attributes of wildfires and the factors affecting **wildfire propagation** … fire **propagation model** in this work. The Wangzhengfei …

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An agent-based **model** to represent space-time **propagation** of **forest-fire** smoke

AK Smith, S Dragicevic - ISPRS Annals of the …, 2018 - isprs-annals.copernicus.org

… -based **model** (ABM) for the **propagation** of **forest-fire** smoke … patterns of spatio-temporal **propagation**. The developed ABM … **model** was developed using data from the 2017 **forest fire** …

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Integrated **wildfire** danger models and factors: A **review**

I Zacharakis, VA Tsihrintzis - Science of the total environment, 2023 - Elsevier

… apply best to fire **behavior prediction**, while other techniques referenced in the current **review** are potentially useful but further investigation is needed. In conclusion, **wildfire** danger is a …

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A **review** of optimization and decision models of prescribed burning for **wildfire** management

J Qi, J Zhuang - Risk Analysis, 2024 - Wiley Online Library

… The selected papers had to meet at least one of the three following criteria: (a) contribute to **wildfire** mitigation, (b) **model** fire **behavior** or influencing factors and their effects on fire behav…

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Fire behaviour and smoke modelling: **model** improvement and measurement needs for next-generation smoke research and forecasting systems

Y Liu, A Kochanski, KR Baker, W Mell… - … of **wildland fire**, 2019 - CSIRO Publishing

… and safety concerns associated with **wildland fire** emissions. This **review** paper presents simulations and … The fire’s rate of **spread** is obtained from the Rothermel **model**. In its simplest …

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Rapid **prediction** of **wildfire spread** using ensemble Kalman filter and polyline simplification

S Yoo, J Song - Environmental Modelling & Software, 2023 - Elsevier

… This paper proposes a new method for rapid **prediction** of **wildfire spread**, which employs computational **wildfire** simulations by FARSITE and assimilates the **simulation** results with …

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Wildfire path **prediction spread** using machine learning

M Makhaba, S Winberg - 2022 International Conference on …, 2022 - ieeexplore.ieee.org

… The methodology of this study started with a literature **review** on the existing ML models in the **wildfire spread** domain, followed by a description of the high-level design of the two …

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Recurrent convolutional deep neural networks for modeling time-resolved **wildfire spread behavior**

J Burge, MR Bonanni, RL Hu, M Ihme - Fire Technology, 2023 - Springer

… To assess the accuracy of the DNN models in predicting the **dynamics** of **wildfire propagation**, we measure the accuracy of each DNN **model** over a time period of 1500 min (100 …

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Some requirements for simulating **wildland fire behavior** using insight from coupled weather—**wildland fire** models

J Coen - Fire, 2018 - mdpi.com

… **Wildland fire behavior** models have been developed and … environmental conditions on fire **behavior**. Early models were … fire **behavior simulation** results to advise potential users on …

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Mathematical modeling of **wildland fire** initiation and **spread**

V Agranat, V Perminov - Environmental Modelling & Software, 2020 - Elsevier

The aim of this paper is to create a user-friendly computational tool for analysis of **wildland fire behavior** and its effect on urban and other structures. A physics-based multiphase …

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